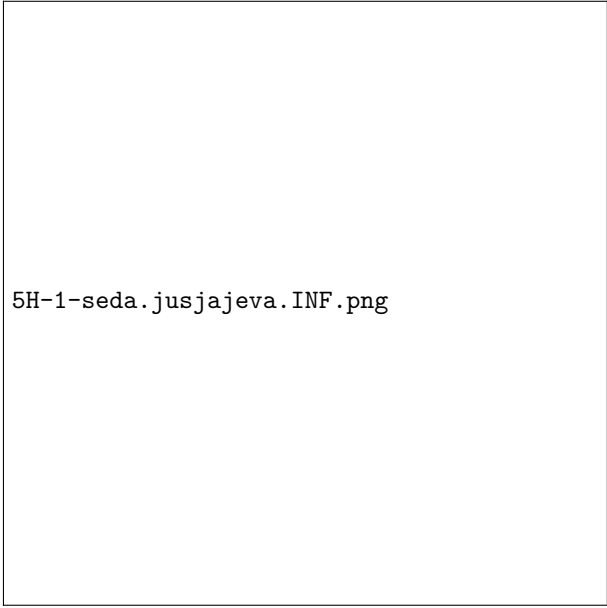


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$$\begin{cases} x(t) = t^2 \\ y(t) = 4t - t^3 \end{cases}$$



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$$\begin{aligned} V &= \pi \int_0^2 (4t - t^3)^2 (t^2)' dt \\ &= \pi \int_0^2 (16t^2 - 8t^4 + t^6) 2t dt \\ &= \pi \int_0^2 (32t^3 - 16t^5 + 2t^7) dt \\ &= \pi \left[8t^4 - \frac{8t^6}{3} + \frac{t^8}{4} \right]_0^2 \\ &= \pi \left(8 \cdot 2^4 - \frac{8 \cdot 2^6}{3} + \frac{2^8}{4} \right) - \pi \cdot 0 \\ &= \frac{64\pi}{3} \end{aligned}$$

References